

some random errors to each measurement. He will also manipulate all the measurements by a fixed amount in order to simulate a handset clock error. He may also send a subset of satellite measurements instead of the complete set of satellites in view.

Architecture of satellite telephone communication interception systems

Monitoring systems are commercially available. These systems log the location of Thuraya handsets operating within user-selected spot beams and the telephone numbers with which they communicate.

Thuraya has a pair of signaling channels. One is transmitted at L-band

Thuraya satellite.

The position of the target is known to be within GPS accuracy (typically 10-20 metres under current conditions). A GIS mapping software is integrated with the system to plot the location of the target set.

Each receiver is independently tunable to any spot beam of the Thuraya network. The interception system monitors L-band link from the mobile to the satellite and C-band feeder link from the satellite to the earth station. Since the power transmitted by the mobile handset on the uplink is very low and so difficult to monitor, the system monitors this information on the C-band downlink (satellite to earth). It automatically

and huge quantities of arms, ammunition and communication equipment recovered from them, how is it that not even a single Thuraya phone of the required modifications has been recovered yet?

There is a good reason as to why the handset cannot be recovered. If it is accepted that Pakistani intelligence agencies are spoofing the Thuraya sets so that their location may not be known and hence the complicity of Pakistan in the affair may be denied, why would they run the risk of giving such sets to the terrorists? They know it well that terrorists may be killed and the phones recovered from them. In that case, spoofed phones would make for an incontrovertible evidence of their complicity, thereby defeating the very purpose of the entire exercise. The risk is not worth the trouble.

There is little which can be done to prevent dynamic spoofing of the LS, if at all it is being done. Spoofing can be detected at the server but getting the original signal is difficult or at least not worth the effort. It therefore makes little sense to cry about it.

All the commercially available interception systems use proprietary technologies. These have therefore not been subjected to peer-reviewed research in which the functioning of their systems and sub-systems is systematically analysed for the possibility of the unintended introduction of such errors. In all probability, the time-variable GPS coordinates which the intelligence agencies are supposedly getting and which they think are due to dynamic spoofing, are to be attributed to hitherto unknown measurement errors or system errors in their interception systems. ●

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by the satellite and received by all Thuraya phones in the spot beam. This signaling channel is called broadcast common control channel (BCCH). In addition to BCCH, access-grant control channel, paging-control channel, frequency-correction channel and basic alerting channel are time-division-multiplexed on the same carrier.

The other signaling channel is called random-access control channel (RACH). It provides access to the network for Thuraya phones. All Thuraya phones within the spot beams transmit bursts at L band on the RACH channel. The satellite relays these bursts to PGW at the C band.

The interception systems typically use a scanning technique to identify C-band downlink frequencies for spot beams of interest. These channels can subsequently be monitored to record the position of mobiles operating within that spot beam and capture details of the telephone numbers that the mobiles contact. The scanning process is performed once in 24 hours, as the frequencies used are dynamically allocated and changed periodically by

maps the uplink RACH control channel and the appropriate TCH channel of the suspect's handset at the C-band to appropriate L-band downlink channel of the target spot beam. After that, it is only a matter of deciphering the cipher algorithms used on Thuraya network.

Spoofing: Is it practical?

Location server spoofing, in any case, is not easy to accomplish as both the hardware and the technical knowledge required are not easy to come by. It is difficult to believe that the terrorists or the Pakistani intelligence agencies have this level of technical know-how. Further, the costs involved also need to be kept in mind. The immense cost would not really be worth the trouble.

In any case, in spite of the popular belief, so far not a single Thuraya set has been recovered that could be stripped and analysed to prove that it is spoofed or its hardware has been modified to mislead the LS.

But given that scores of terrorists have been killed by the security forces

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